Five-backend Standard Model timings

Descriptive shared-host measurements. These full-amplitude timings compare explicit CutTools, Ninja, COLLIER, NinjaRS and TIRust on the selected saved cohorts. There is no CPU reservation and no confidence-interval or final local-performance acceptance claim. The earlier 39 pages retain their separately bound scientific and timing history.

GNU Fortran (GCC) 15.2.0; AMD EPYC 9754 128-Core Processor; affinity CPU 12; one-thread environment on a shared, unreserved host.

Elapsed time is the unchanged Fortran `SYSTEM_CLOCK` wall clock, divided by the actual `points × loops`. Each backend has ten position-balanced cycles (five Latin positions followed by reversal), with loops fixed after two retained calibrations. The nominal batch target is two seconds; actual batch durations and counts remain in the closed evidence. The table reports the median and **unscaled** median absolute deviation, not an error bar on the mean. Ratios are medians of ten within-cycle backend / NinjaRS8 ratios. Actual timed batches range from 1.95 to 3.14 seconds; integer full-cohort traversal counts can exceed the nominal target.

Table 1: Warmed full-amplitude wall time. Ratios below one mean that backend is faster than NinjaRS8. All actual saved-cohort sizes and accepted output-call counts are retained.

Process · cohort	Reducer	READY s	Wall ms/point · median ± MAD	Ratio / 8	Accepted DP / QP	Status
$u\bar{u} \to Z$ · QCD N=100	CutTools	0.0343	$0.127 \pm 9.53 \times 10^{-4}$	×1.48	157000 / 0	3e-6 pass
	Ninja	0.0282	$0.0876 \pm 9.14 \times 10^{-4}$	×1.01	237000 / 0	3e-6 pass
	COLLIER	0.0358	$0.116 \pm 7.30 \times 10^{-4}$	×1.37	173000 / 0	3e-6 pass
	NinjaRS	0.0297	$0.0854 \pm 1.78 \times 10^{-3}$	×1.00	239000 / 0	3e-6 pass
	TIRust	0.0166	$0.0946 \pm 1.36 \times 10^{-3}$	×1.11	214000 / 0	3e-6 pass
$gg \to t\bar{t}$ · QCD N=100	CutTools	0.615	5.55 ± 0.0559	×1.30	4000 / 0	3e-6 pass
	Ninja	0.447	4.23 ± 0.0255	×0.998	5000 / 0	3e-6 pass
	COLLIER	0.583	5.40 ± 0.0319	×1.27	4000 / 0	3e-6 pass
	NinjaRS	0.453	4.25 ± 0.0325	×1.00	5000 / 0	3e-6 pass
	TIRust	0.482	4.46 ± 0.0249	×1.05	5000 / 0	3e-6 pass
$e^+e^- \to \mu^+\mu^-$ · EW N=100	CutTools	0.442	4.12 ± 0.0144	×1.21	5000 / 0	3e-6 pass
	Ninja	0.365	3.31 ± 0.0357	×0.975	6000 / 0	3e-6 pass
	COLLIER	0.488	4.49 ± 0.0374	×1.33	5000 / 0	3e-6 pass
	NinjaRS	0.376	3.38 ± 0.0155	×1.00	6000 / 0	3e-6 pass
	TIRust	0.366	3.54 ± 0.0236	×1.04	6000 / 0	3e-6 pass
$u\bar{u} \to e^+e^-\mu^+\mu^-$ · QCD N=100	CutTools	1.53	14.9 ± 0.0813	×1.03	2000 / 0	3e-6 pass
	Ninja	1.47	14.4 ± 0.127	×1.000	2000 / 0	3e-6 pass
	COLLIER	1.55	15.0 ± 0.174	×1.05	2000 / 0	3e-6 pass
	NinjaRS	1.48	14.4 ± 0.0872	×1.00	2000 / 0	3e-6 pass
	TIRust	1.49	14.5 ± 0.0970	×1.01	2000 / 0	3e-6 pass

The separate ordinary 1e-8 target passes in 20 / 20 lanes.

READY time includes the first amplitude and initialization/learning, excluding SETPARA and process startup; it is separate from warmed timing. Accepted DP/QP counts cover timed output calls and do not recover unobserved internal retries or floating-point exception flags. Numerical checks and timing status are separate; failed or unavailable comparisons remain explicit. Coefficient checks cover the retained final 100-point traversal of learning, each calibration and each timed batch; accepted-status counts cover all timed repetitions.

The $u\overline{u} \to Z$ cohort varies physical on-shell Z boosts. The other three selected cohorts use 1 TeV. These labels follow actual saved-point physics and generation records, not the global configuration energy field.

Timing comparison and scope

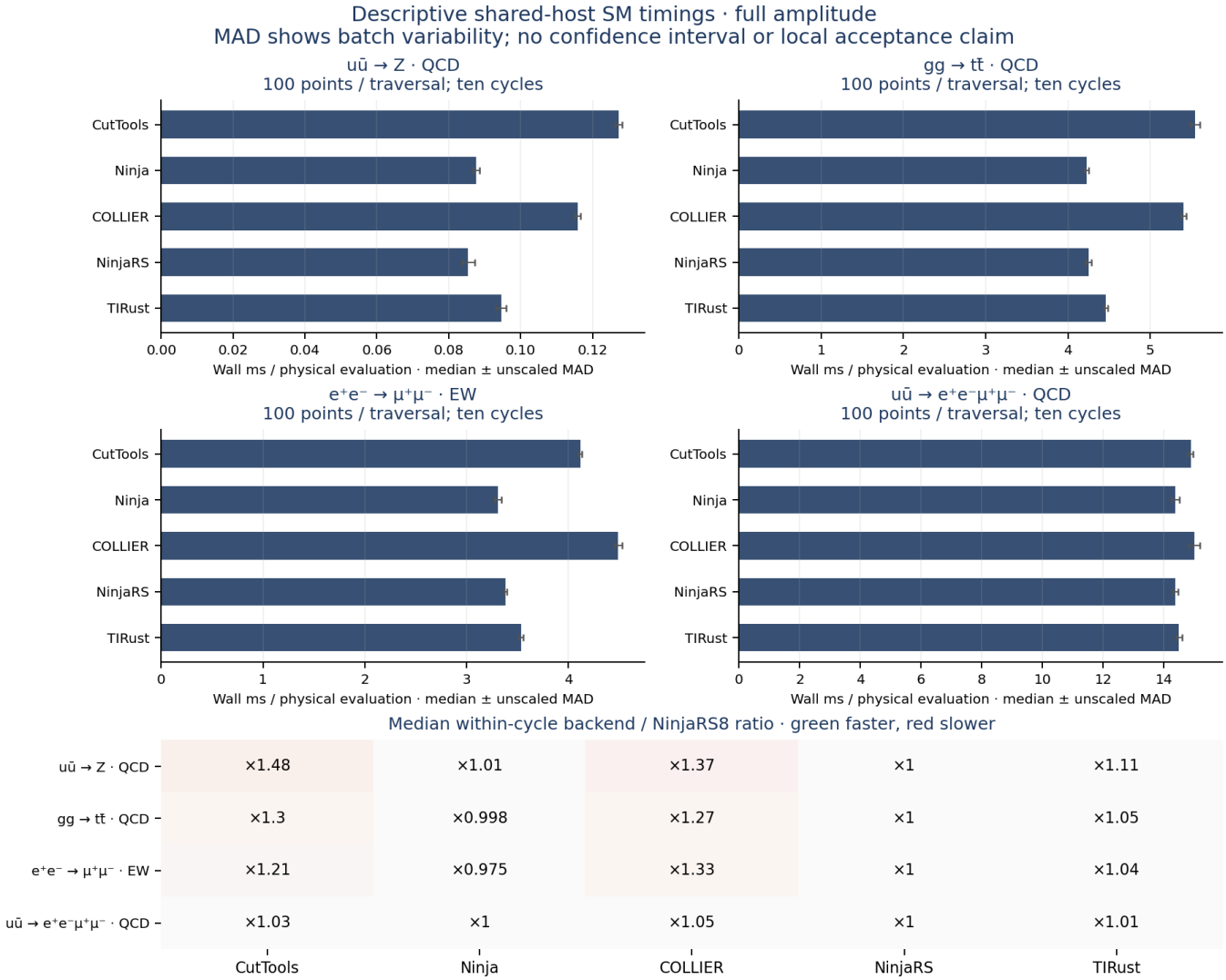


Figure 1: Absolute warmed wall timing for each case and within-cycle ratios relative to NinjaRS8. Whiskers show unscaled MAD, not confidence intervals. Green means a smaller backend/NinjaRS8 ratio, red a larger ratio. Unavailable cells are not zero.

The fresh four-lepton QCD $2 \to 4$ generation has maximum loop N=4 and rank=3, across 12 loop-call groups, as reparsed from its compiled source text. Four final particles do not imply six propagators in each loop. The separately validated ZZgg process retains its full numerical coverage and precise local timing TODO. These shared-host comparisons do not replace the full registered numerical matrix or Hua-Sheng's precise local per-process timing campaign. The agreed upper-95% runtime-ratio bound ≤1.05, source-matched baseline/rescue-chain comparison and existing-backend regressions remain separate acceptance items.

Separate private checked-CMS20 result. One later private CMS20 fixture with cache disabled passes automatic stability checks and genuine QP rescue (return323): six failed DP group reductions remain retained; all52 QP reductions succeed. The earlier failed controller attempt remains failed. This is not generic or production qualification, enabled-cache qualification, independent full-amplitude truth, or a performance result. Closed evidence. This does not qualify the production timing inputs or generic execution.

Closed timing evidence and reproduction commands: Closed timing capsules; see the accompanying README and scientific-selection.json. Clock, source/runtime/card/cohort pins, actual durations, calibration and all accepted-status/precision histograms are retained with that packet.